

Spreading 50 Points in the Plane

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Abstract

We place 50 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 47.93329+. Erich Friedman’s table of best known configurations lists n up to 31. There is no larger configuration in the table to compare with. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 50 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 47.93329\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 50$ is not on the page. There is no larger configuration in the table to compare with.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n50.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
9586659079901342879593792
denominator:
199999999999616427203441
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

50 points, x and y per line. File: friedman-maxmin2d/coords/n50.txt.

```
-10.998123692682 6.285327476891
-9.198957439186 5.857165371096
-6.379312262725 7.954572903163
-7.647843361191 4.594663182585
-6.355240954661 11.106454280088
-7.327758428232 7.637634626627
-6.742458035883 8.886305239962
-8.636344415283 10.552499590006
-7.358813433228 6.638116949612
-9.462745864626 9.989418206434
-10.834415651816 10.085665047285
-4.747978466447 6.785880540979
-6.695199775616 4.898752312386
-9.271351039586 11.325006280651
-8.266555049896 6.218587263690
-9.272162965211 8.938705769594
-8.491061178639 9.563109484360
-8.284837308742 11.488684803546
-9.585644937793 7.989111617934
-5.958387480058 9.506976943431
-5.595241706899 8.575244606632
-10.373553421838 7.373319265944
-9.363558713597 4.870805183100
-11.246296654621 9.174427405652
-8.427045096306 5.221436391962
-5.529232065396 7.427919641415
-5.796300695661 5.336908042756
-7.491157717550 9.549214575145
-4.615023036388 8.773162160126
-5.634806934213 6.323781775821
-11.500607044098 8.207304752725
-4.590826313663 7.773454943682
-11.361269465841 7.217059813690
-7.494642707016 5.582858284556
-10.097752488928 10.761924897079
-5.533589522550 10.536463995865
-10.198706125972 5.420831363935
-4.978168809546 9.704894496925
-7.690904201391 8.569366963426
-7.636440954194 10.538604680791
-6.410367267721 6.955055226149
-7.284933847653 11.474789894332
-8.598645818058 8.149837277503
-9.222499164634 7.057379281135
-10.199409027513 9.313158356640
-10.010407648679 6.441586929145
-6.706748061202 10.170269066548
-6.571861029169 5.968181493084
-10.512891000095 8.363564204980
-8.235500044900 7.218104940704
```